VICTORY 2027

ELECTRICITY AND MAGNETISM

LECTURE-08 PHYSICS

Shailendra sir



Topics To Be Covered

1. ELECTROMAGNETIC INDUCTION
2. FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION
3. AC GENERATOR
4. QUESTIONS

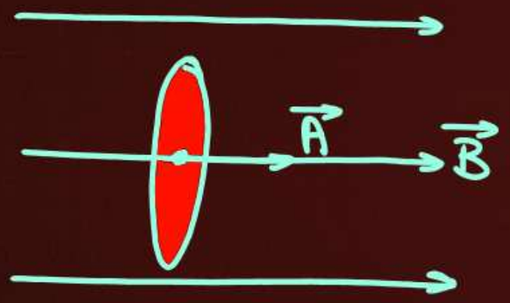

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Revision

Magnetic Flux

$\phi = BA\cos\theta$



Faraday's Law

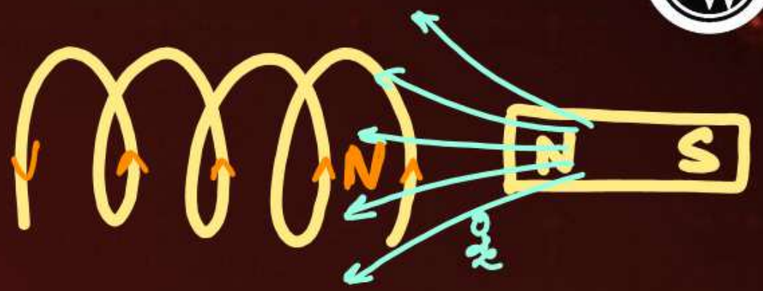
First : $\Delta\phi_B \longrightarrow \mathcal{E}_{in} \longrightarrow I_{in}$

Second : $\mathcal{E}_{in} = \frac{\Delta\phi_B}{\Delta t}$

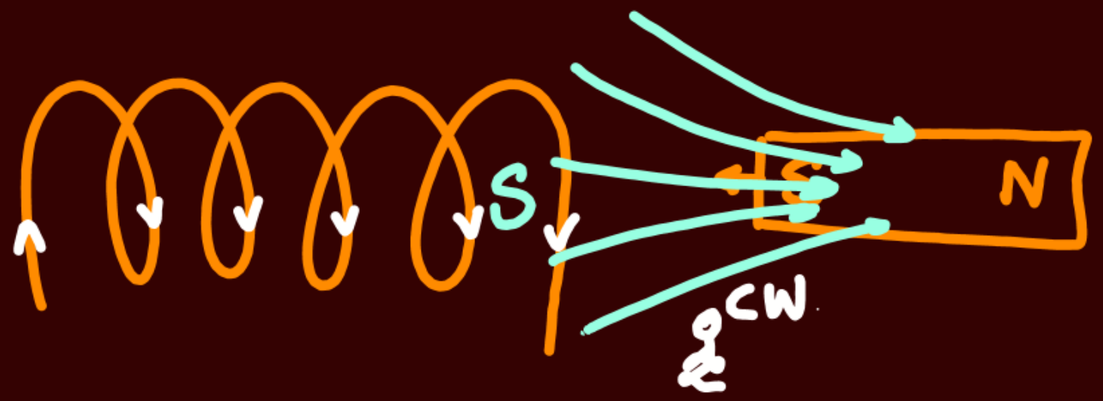
Lenz Law

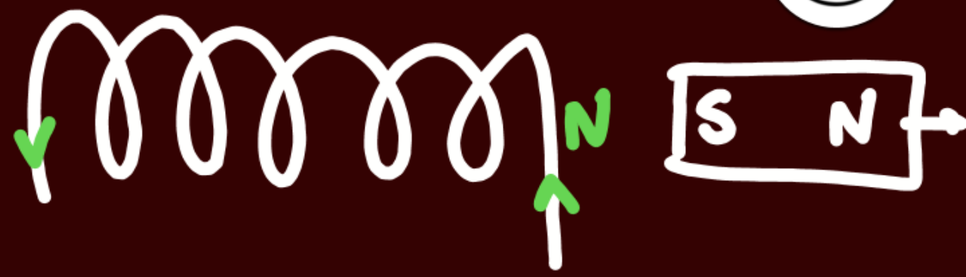
Direction of induced current is always such as to oppose the cause which produce it.

I_{in} → Namak haraam






S
S
N
CW


N
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S
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S



ELECTROMAGNETIC INDUCTION

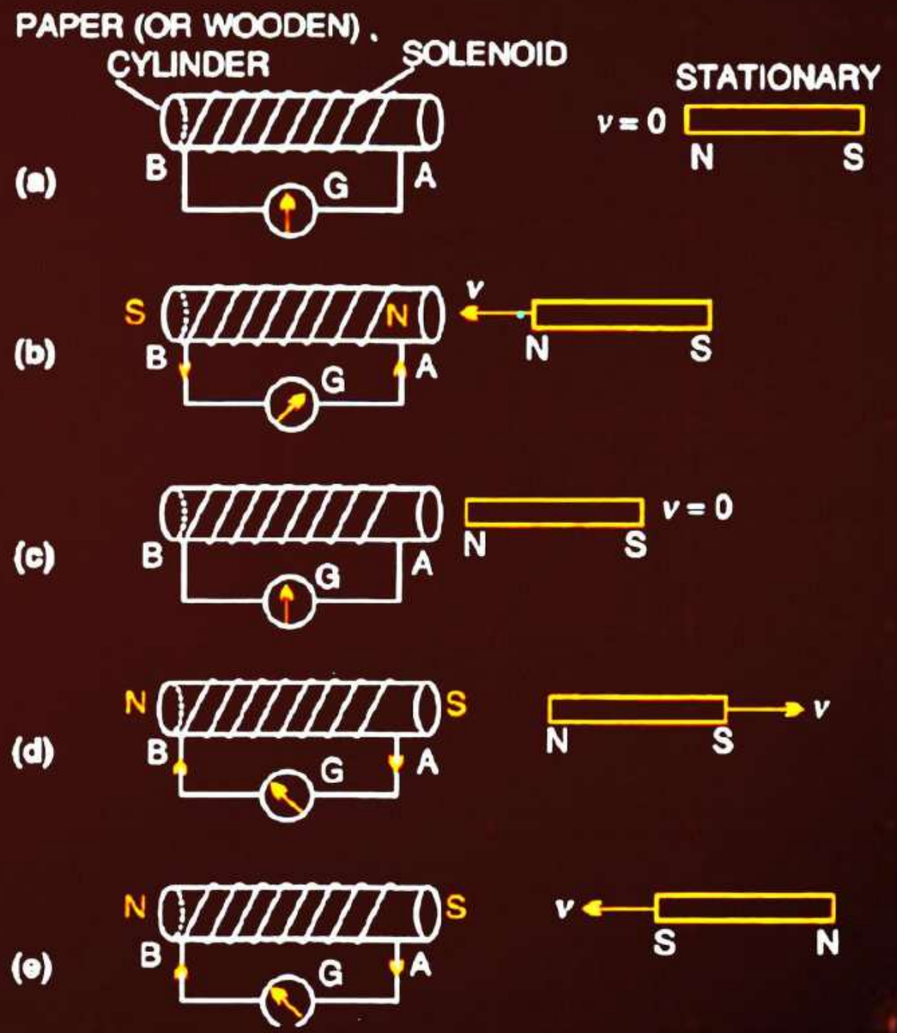
- **Whenever there is a change in the number of magnetic field lines linked with a conductor, an electromotive force (e.m.f.) is developed** between the ends of the conductor which lasts **as long as there is a change in the number of magnetic field lines**. This phenomenon is called the electromagnetic induction.

$$\Delta\phi_B \longrightarrow \mathcal{E}_{in} \longrightarrow I_{in}$$

Demonstration of the phenomenon of electromagnetic induction

Observations:

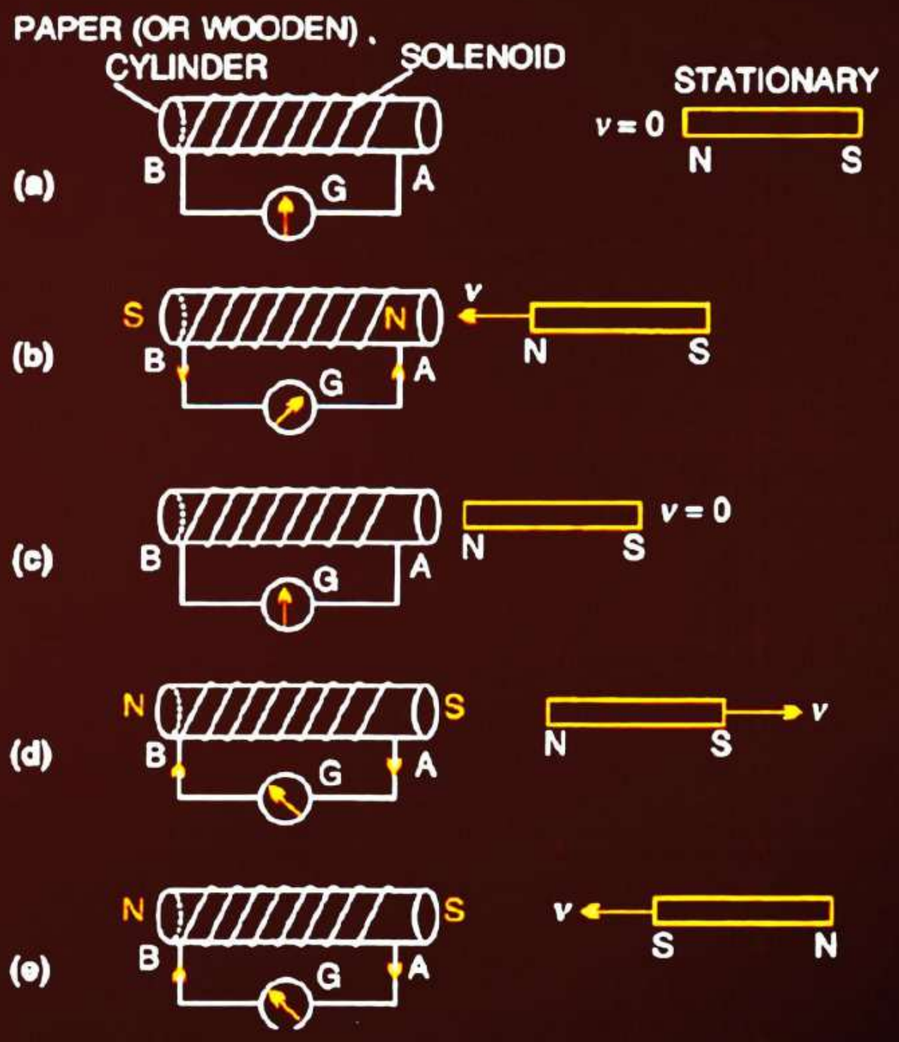
1. **Magnet Stationary (v = 0):**
   - When the **magnet is not moving**, the **galvanometer shows no deflection.**
   - ⇒ **No current** is induced in the solenoid.
2. **Magnet Moved Towards Solenoid (North Pole Facing):**
   - **Galvanometer deflects to the right**, showing **current flows from B → A.**
   - ⇒ **Current is induced** as the **magnet moves** towards the solenoid.
3. **Magnet Motion Stops:**
   - When the **magnet stops, pointer returns to zero.**
   - ⇒ **Induced current flows only while magnet is moving.**
4. **Magnet Moved Away from Solenoid:**
   - **Galvanometer deflects to the left**, showing **current flows from A → B.**
   - ⇒ **Induced current reverses direction** when the **magnet moves away.**



Demonstration of the phenomenon of electromagnetic induction

- If the magnet is moved away rapidly, the deflection in the galvanometer increases although the direction of deflection remains same. It shows that now more current flows.

- If the magnet is brought towards the solenoid by keeping its south pole towards it, the pointer of galvanometer deflects towards left [Fig. (e)] showing that the current in solenoid flows in direction A to B which is opposite to that shown in Fig. (b).



Demonstration of the phenomenon of electromagnetic induction

Exactly similar observations are made in galvanometer if magnet is kept stationary and the solenoid is moved. It implies that it is the relative motion between the solenoid and magnet which produces a current, irrespective of whether the magnet moves or the solenoid moves.

CONCLUSIONS:

- A current flows in the coil only when there is a relative motion between the coil and magnet.
- The direction of current is reversed if the direction of motion (or polarity of the magnet) is reversed.
- The current in the coil is increased
  - (i) by the rapid motion of magnet (or coil),
  - (ii) by the use of a strong magnet,
  - (iii) by increasing the area of cross section of coil and by increasing the number of turns in the coil.

$$\mathcal{E}_{in} = \frac{\Delta\phi}{\Delta t} = \frac{N\Delta(BA)}{\Delta t}$$

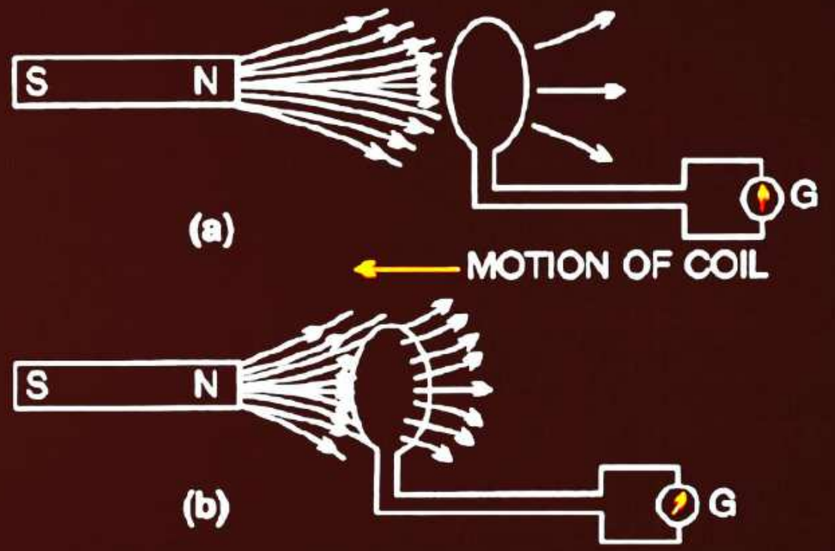
Faraday's Explanation

- **No relative motion**: Magnetic flux through the coil remains constant → no e.m.f. and no current.

- **Relative motion**: Magnetic flux linked with the coil changes → e.m.f. is induced.

- Moving the coil towards the magnet → flux increases.
- Moving the coil away from the magnet → flux decreases.

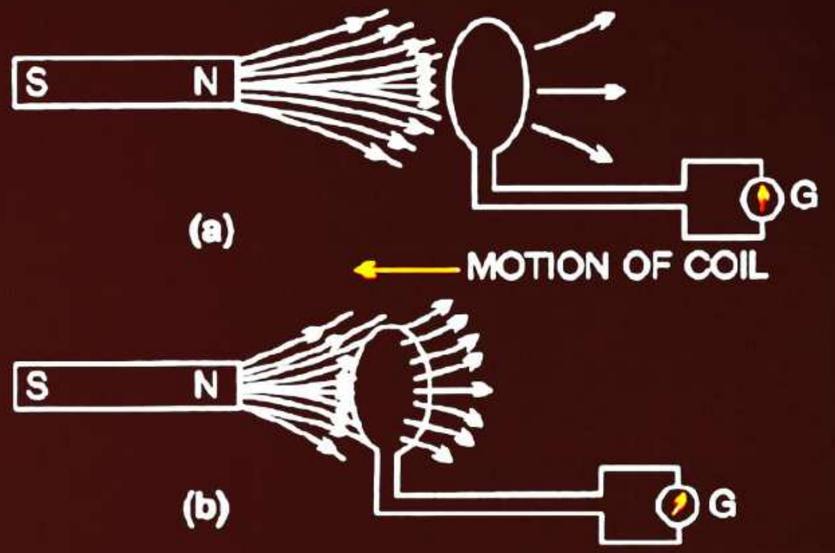
✓ If the coil circuit is closed, the induced e.m.f. produces current.



Faraday's Explanation

Electromagnetic induction is the phenomenon in which an e.m.f. is induced in the coil if there is a change in the magnetic flux linked with the coil.

In electromagnetic induction, the mechanical energy spent (i.e., the work done) in moving the coil or the magnet, to change the magnetic flux is transformed into the electrical energy in the form of current in the coil.



FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION

On the basis of the above experimental observations, Faraday formulated the following two laws of electromagnetic induction :

$$\Delta\phi_B \longrightarrow \mathcal{E}_{in} \longrightarrow I_{in}$$

(1) **Whenever there is a change in magnetic flux linked with a coil, an e.m.f. is induced. The e.m.f. induced lasts so long there is a change in magnetic flux linked with the coil.**

(2) **The magnitude of e.m.f. induced is directly proportional to the rate of change of magnetic flux linked with the coil. If magnetic flux changes at a constant rate, a steady e.m.f. is produced.**

$$\mathcal{E}_{in} = \frac{\Delta\phi_B}{\Delta t}$$

If the circuit of coil is closed, a current flows in it due to the e.m.f. induced across its ends.

FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION

The magnitude of e.m.f. induced in a coil is equal to the product of rate of change of magnetic flux linked with each turn of the coil and the number of turns in the coil.

$$\textbf{Induced e.m.f} = \frac{\text{change in magnetic flux of each turn} \times \text{number of turns in the coil}}{\text{time in which the magnetic flux changes}}$$

$$\mathcal{E}_{in} = N\frac{\Delta\Phi_B}{\Delta t}$$

Thus, for a given coil and magnet, it depends on the following two factors :

(1) on the rate of change of magnetic flux with each turn, and
(2) on the number of turns in the coil.

$$\mathcal{E}_{in} = \frac{\Delta\phi_B}{\Delta t}$$

$$I_{in} = \frac{\mathcal{E}_{in}}{R} = \frac{1}{R}\frac{\Delta\phi_B}{\Delta t}$$

Direction : Flemming's Right Hand Rule.

Fore → $\vec{B}$

Thumb → $\vec{F}$ or Motion of Conductor

Middle Finger → I_{in}

$$\frac{\Delta q}{\cancel{\Delta t}} = \frac{1}{R}\frac{\Delta\phi_B}{\cancel{\Delta t}}$$

$$\Delta q = \frac{\Delta\phi_B}{R}$$

Flemming's

left hand Rule.

Dir. of Force

Fore → $\vec{B}$

Middle → I

Then Thumb → $\vec{F}$

Right Rule

Dir. of induced current

Fore → $\vec{B}$

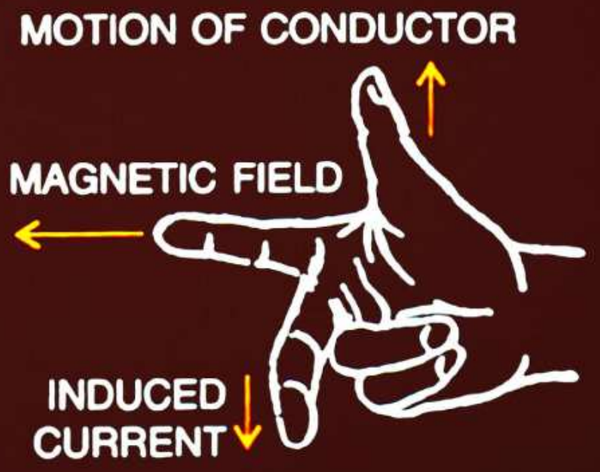
Thumb → Motion

Then middle → Iin

DIRECTION OF INDUCED E.M.F.

(1) Fleming's right hand rule

Stretch the thumb, central finger and forefinger of your right hand mutually perpendicular to each other as shown in Fig. 10.31. If the forefinger indicates the direction of magnetic field and the thumb indicates the direction of motion of the conductor, then the central finger will indicate the direction of induced current.



Thumb—Motion
Fore finger—Field
Central finger—Induced current



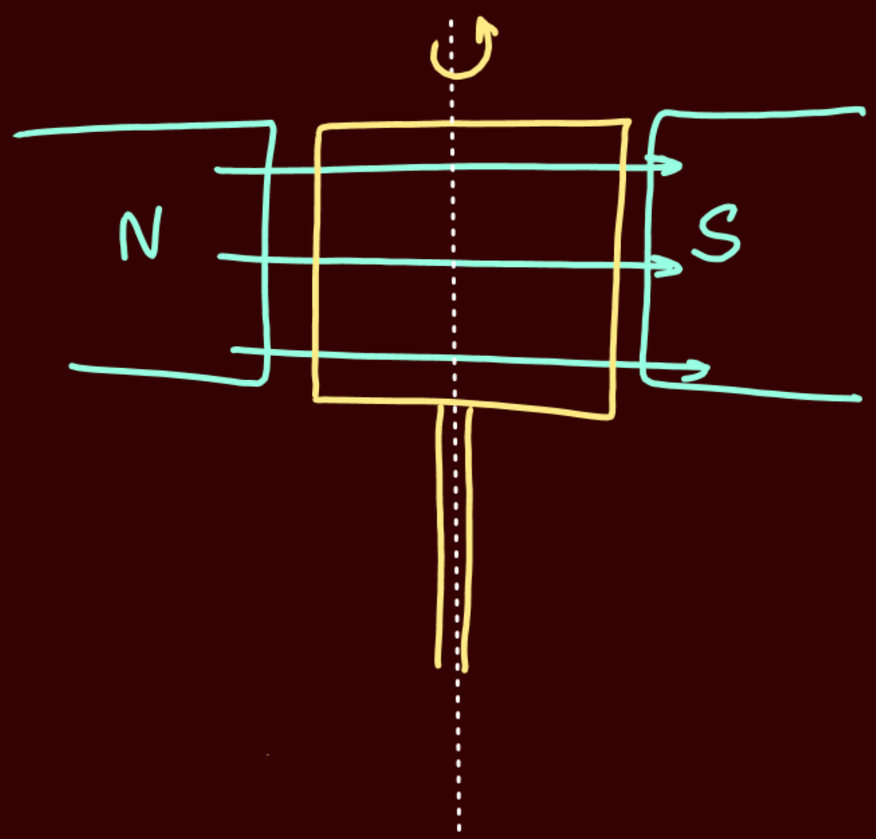
DIRECTION OF INDUCED E.M.F.

(2) Lenz's Law:

According to Lenz's law, the direction of induced e.m.f. (or induced current) is such that it opposes the cause which produces it.

Lenz's law is based on the law of conservation of energy. It shows that the mechanical energy spent in doing work against the opposing force experienced by the moving magnet, is transformed into the electrical energy due to which current flows in the solenoid.

$\Phi_B = BA\cos\theta$





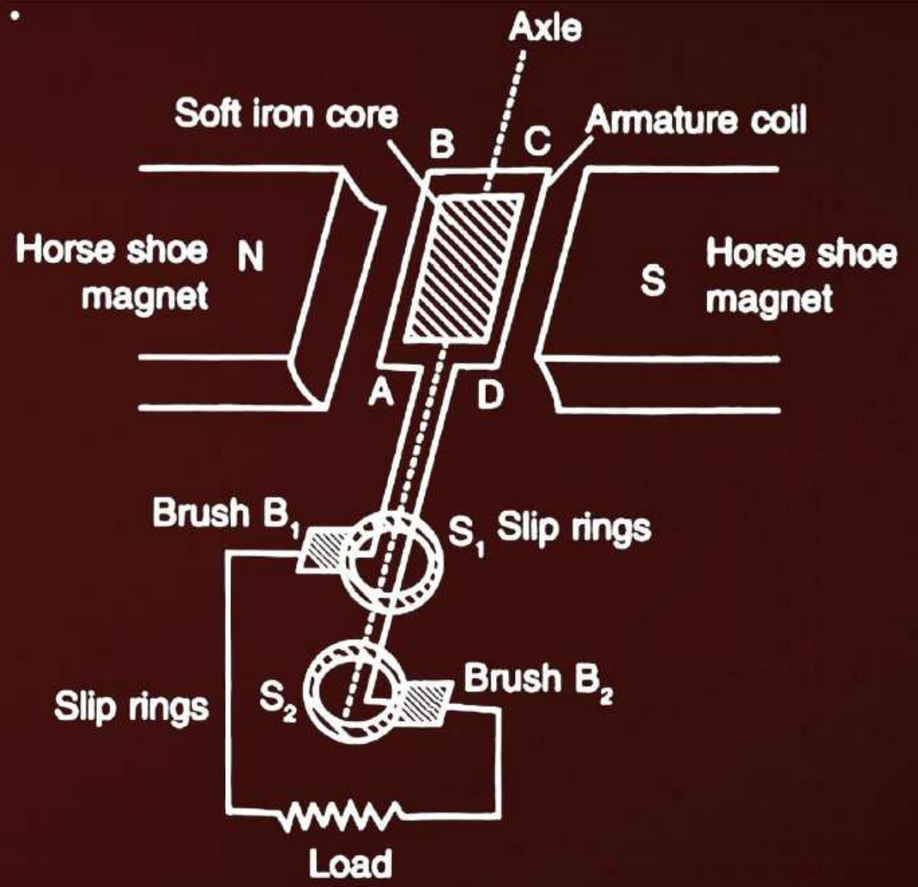
A.C. GENERATOR

- An a.c. generator is a device which converts the mechanical energy into the electrical energy using the principle of electromagnetic induction.

- **Principle:** In a generator, a coil is rotated in a magnetic field. Due to rotation, the magnetic flux linked with the coil changes and therefore an e.m.f. is induced between the ends of the coil. Thus a generator acts as a source of current in an external circuit containing load when it is joined between the ends of its coil.

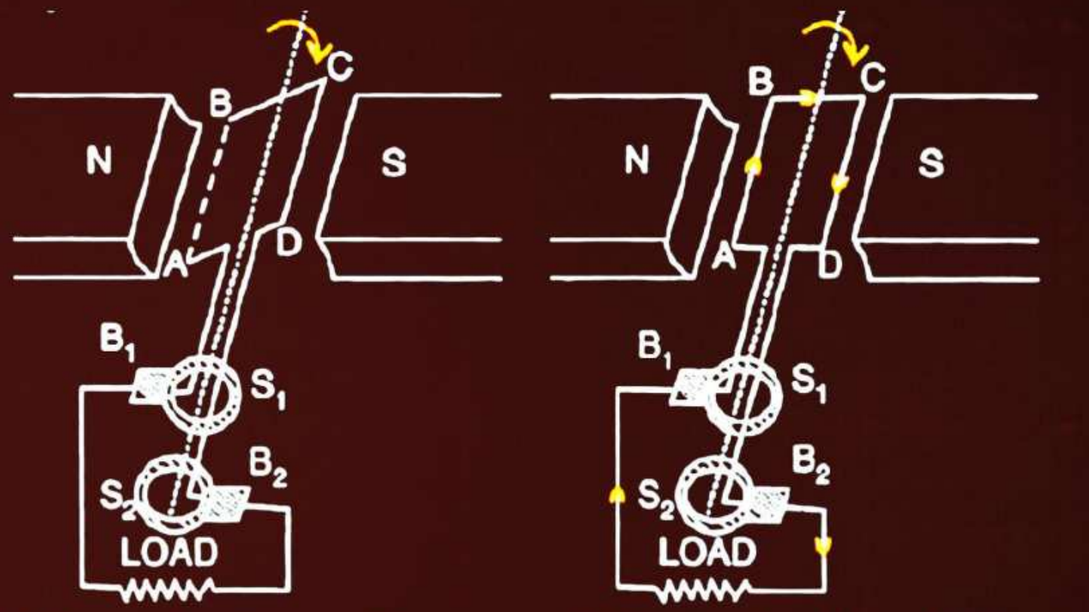
Construction and its main parts

- **Horseshoe Electromagnet** (N, S): Produces a strong magnetic field between its poles.
- **Armature Coil** (ABCD): A large number of insulated copper-wire turns wound on a soft iron core. It rotates rapidly in the magnetic field.
- **Slip Rings** (S_1, S_2): Two metallic rings connected to the two ends of the coil. They rotate with the coil and maintain contact with the brushes.
- **Brushes** (B_1, B_2): Carbon brushes that press gently against the slip rings. They do not rotate and act as terminals for connecting the external load.



Working of AC Generator

- Initially, the coil is perpendicular to the magnetic field ($\theta = 0°$), so maximum magnetic flux passes through it.
- When the coil is at rest, no e.m.f. is induced.
- When the coil is rotated clockwise, the magnetic flux through it decreases, inducing an e.m.f. and current.
- By Fleming's Right-Hand Rule, the induced current in the coil flows A → B → C → D.
- Hence, in the external circuit, current flows from B_2 → load → B_1.

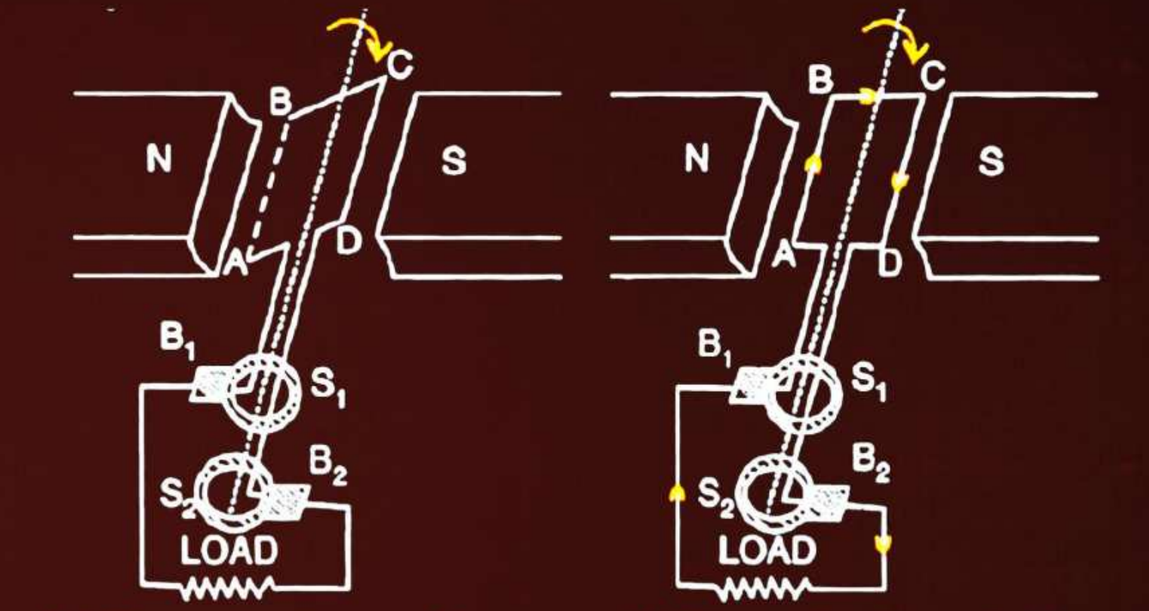


(a) Plane of coil normal to magnetic field (initial position)

(b) Plane of coil parallel to magnetic field (after 90° rotation)

Working of AC Generator

- At $\theta = 90°$, the plane of the coil becomes parallel to the magnetic field.
- Therefore, magnetic flux through the coil = 0.
- At this position, the rate of change of magnetic flux is maximum.
- Hence, the induced e.m.f. is maximum.

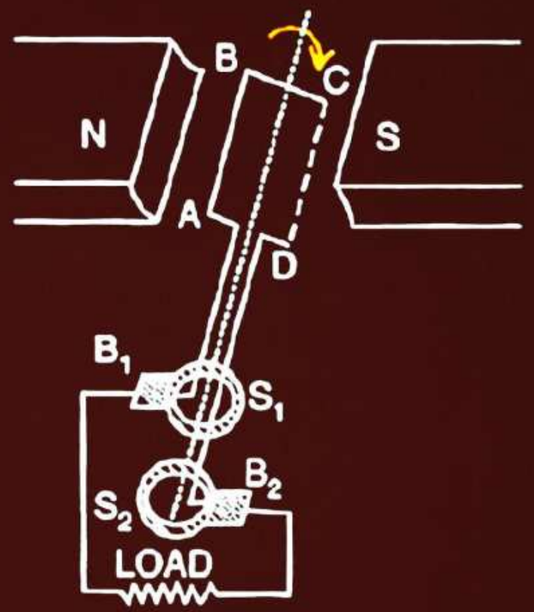


(a) Plane of coil normal to magnetic field (initial position)

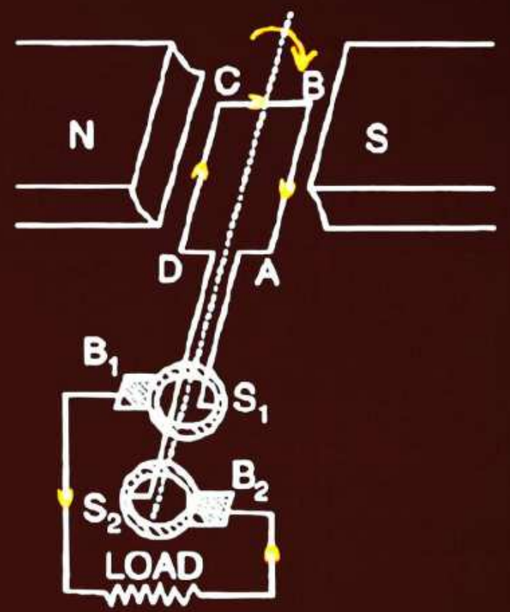
(b) Plane of coil parallel to magnetic field (after 90° rotation)

Working of AC Generator

- At $\theta = 180°$: The coil is again perpendicular to the magnetic field, so magnetic flux is maximum.
- The rate of change of flux is zero, hence induced e.m.f. = 0.
- On further rotation, e.m.f. increases again but in the opposite direction.
- At $\theta = 270°$: The rate of change of magnetic flux is maximum, so induced e.m.f. is maximum in the opposite direction.
- The positions of arms AB and CD are interchanged, so the current in the coil flows D → C → B → A.
- Therefore, in the external circuit, current flows B_1 → load → B_2.



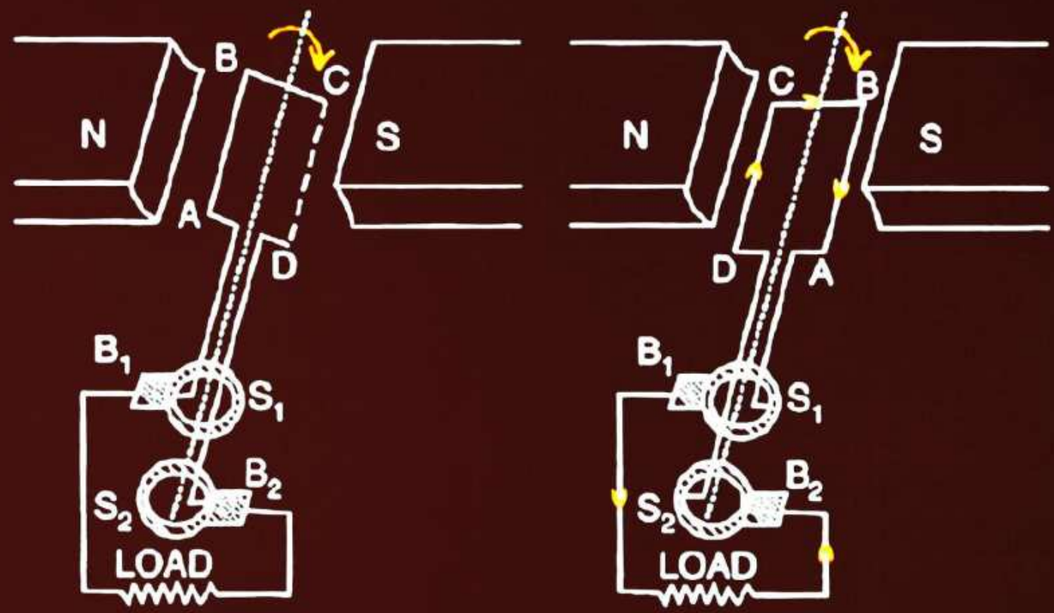
(c) Plane of coil normal to magnetic field (after 180° rotation)



(d) Plane of coil parallel to magnetic field (after 270° rotation)

Working of AC Generator

- At $\theta = 360°$, the coil again becomes perpendicular to the magnetic field.
- Magnetic flux becomes maximum, while the rate of change of flux is zero.
- Therefore, the induced e.m.f. becomes zero.
- During each complete rotation, the induced e.m.f. changes both its magnitude and polarity.
- Hence, it is called an alternating e.m.f.
- This produces alternating current (AC) in the external circuit. The direction of current reverses after every half rotation (180°).



(c) Plane of coil normal to magnetic field (after 180° rotation)

(d) Plane of coil parallel to magnetic field (after 270° rotation)

Working of AC Generator

Fig. represents the variation in e.m.f. induced between the ends of coil with angle θ which the normal to the plane of coil makes with the magnetic field.

$$\mathcal{E} = NBA\omega \sin \omega t$$

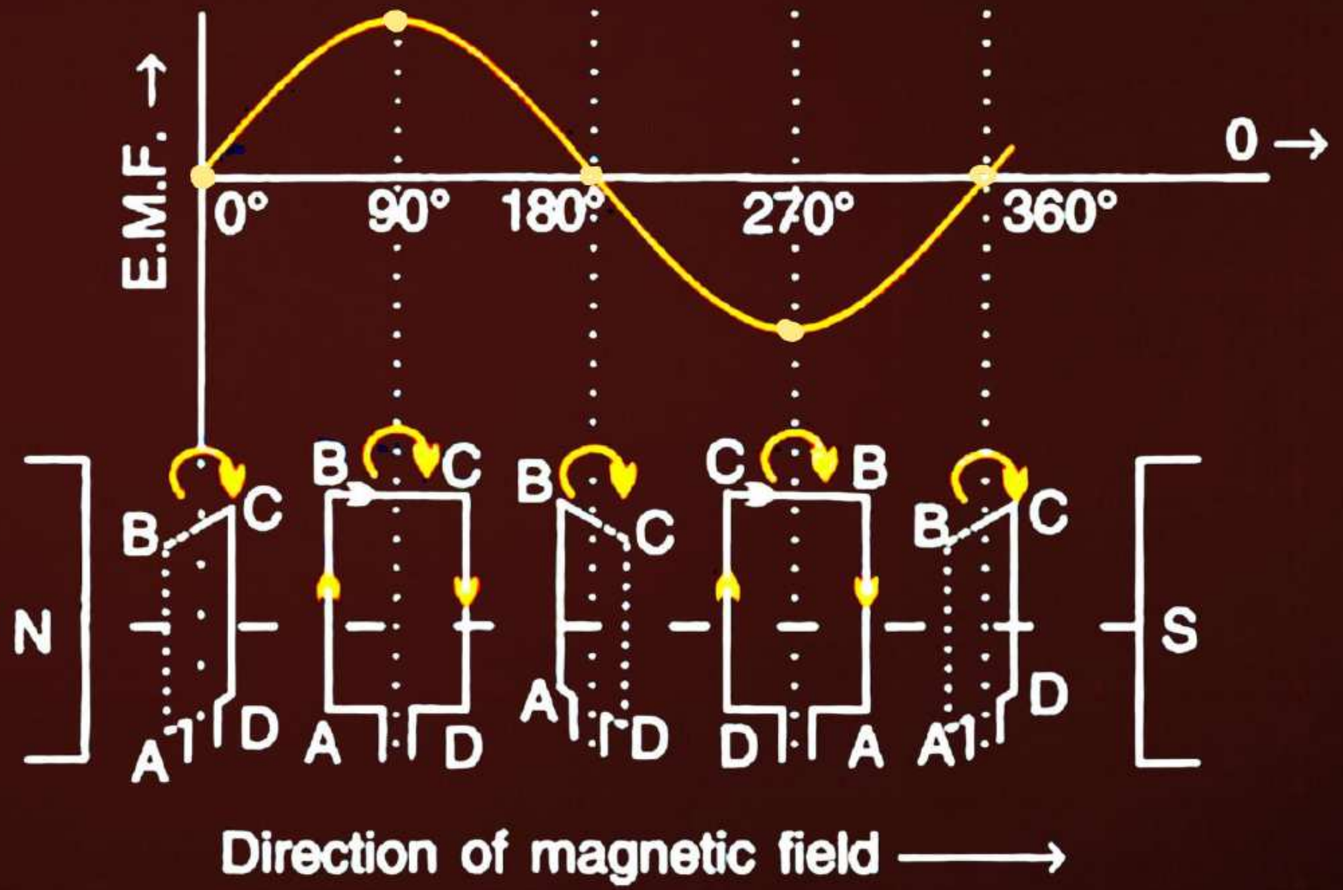
$$\Downarrow$$

$$\mathcal{E} = \mathcal{E}_0 \sin \omega t$$

Peak EMF

$$\mathcal{E}_0 = NBA\omega$$

$$\omega = 2\pi f$$





Note

$\mathcal{E}_{in} \propto \Delta\phi_B$

(1) The magnitude of induced e.m.f. becomes maximum when the magnetic flux linked with coil reduces to zero from its maximum value. This happens when the plane of coil lies in direction of the magnetic field.

(2) The induced e.m.f. becomes zero when the magnetic flux linked with coil is maximum. This happens when the plane of coil is normal to the magnetic field.

HOMEWORK

- REVISE THIS LECTURE AGAIN.
- PRACTISE 20 QUESTIONS DAILY.
- COMMENT YOUR RESPONSE ABOUT THIS CLASS.



QUESTIONS

State with reason in a d.c. motor, the effect of

(i) inserting a soft iron core within the coil,

(ii) increasing the area of the coil, and

(iii) increasing the strength of current in the coil.

(i) On inserting a soft iron core within the coil of a d.c. motor, the speed of rotation of coil increases. The reason is that on inserting the soft iron core in the coil, the strength of magnetic field between the pole pieces of magnet increases due to which the deflecting couple on the coil increases.

(ii) On increasing the area of coil, the deflecting couple on the coil increases, so the speed of rotation of coil increases.

(iii) By increasing the strength of current in the coil, the deflecting couple on the coil increases, so the speed of rotation of the coil increases.

Thank You